

Index laws



1 Write the following expressions in simplest index form:

a $2^{12} \times 2^9$

b $2^8 \times 11^8$

c $11^{12} \div 11^8$

d $21^5 \div 3^5$

e $(5^{12})^4$

f $15^{17} \div 15^8 \div 15^5$

g $(23^8)^9 \times 23^7$

h $\frac{(17^5)^8}{17^{32}}$

i $\frac{19^9 \times 19^4}{19^8}$

j $\frac{12^6}{12^4} \times 12^5$

k $\frac{(13^5)^2 \times 13^3}{13^5}$

l $\frac{(15^9)^5 \times 15^7}{15^{25}}$

2 Complete the following statements:

a $11^{11} \times 11^{\square} = 11^{19}$

b $5^{11} \times (\square)^{11} = 35^{11}$

c $97^{\square} \div 97^{22} = 97^{12}$

d $55^6 \div (\square)^6 = 11^6$

e $(11^4)^{\square} = 11^{12}$

f $7^8 \times 7^{\square} = 7^{14}$

g $2^{10} \times (\square)^{10} = 10^{10}$

h $19^{\square} \div 19^{18} = 19^{20}$

i $60^4 \div (\square)^4 = 12^4$

j $(13^8)^{\square} = 13^{16}$

3 Evaluate the following expressions:

a $6^5 \times 6^3$

b $7^3 \times 3^3$

c $4^8 \div 4^3$

d $(5^4)^2$

e $35^5 \div 5^5$

f $2^4 \times 4^4$

g $11^{18} \div 11^9 \div 11^7$

h $(3^3)^2$

i $\frac{6^5 \times 6^9}{6^{12}}$

j $7^{27} \div 7^{30} \div 7^3$

k $\frac{12^{10} \times 12^4}{12^{11}}$

l $\frac{(6^8)^6}{6^{46}}$

Negative bases

4 Write the following expressions in simplest index form:

a $(-11)^{10} \times (-11)^3$

b $(-7)^8 \times 3^8$

c $(-5)^2 \times 3^2$

d $(-3)^{12} \div (-3)^5$

e $(-12)^{20} \div (-12)^{19}$

f $(-30)^{50} \div (-30)^{47}$

g $(-48)^3 \div (-6)^3$

h $(-33)^{11} \div (-3)^{11}$

i $(-35)^5 \div 5^5$

j $(-42)^2 \div 7^2$

5 Complete the following statements:



a $11^3 \times (\square)^3 = (-77)^3$

b $(-5)^{11} \times (\square)^{11} = 15^{11}$

c $(-5)^{\square} \div (-5)^{31} = (-5)^{19}$

d $(-14)^{13} \div (\square)^{13} = (-7)^{13}$

e $(-3)^{\square} \div (-3)^{39} = (-3)^{10}$

f $(-5)^4 \times (\square)^4 = (-60)^4$

g $(-3)^7 \times (\square)^7 = (6)^7$

h $(\square)^3 \div (-3)^3 = (7)^3$

i $(-33)^3 \div (\square)^3 = -11^3$

j $(\square)^7 \div (-5)^7 = 11^7$

6 Evaluate the following expressions:

a $(-4)^{11} \div (-4)^7$

b $(-2)^3 \times (-2)^3$

c $(-3)^3 \times (-3)^2$

d $4^3 \times (-5)^3$

e $(-3)^8 \div (-3)^5$

f $15^5 \div (-3)^5$

g $2^3 \times (-3)^3$

h $(-14)^{11} \div 2^{11}$

i $(-7)^2 \times 5^2$

j $(-9)^4 \times (-3)^4$

k $(-100)^6 \div 50^6$

l $60^3 \div (-3)^3$

Fractional bases

7 Write the following in simplest index form:

a $\left(\frac{1}{3}\right)^4$

b $\left(\frac{3}{8}\right)^3$

c $\left(\frac{4}{16}\right)^8$

d $\left(\frac{15}{6}\right)^2$

e $\left(\frac{10}{33}\right)^5$

f $\left(\frac{2}{35}\right)^6$

g $\left(\frac{5}{18}\right)^3$

h $\left(\frac{29}{41}\right)^7$

i $\left(\frac{11}{13}\right)^9$

j $\left(\frac{20}{3}\right)^2$

k $\left(\frac{17}{4}\right)^4$

l $\left(\frac{31}{50}\right)^5$

8 Complete the following statements:

a $\frac{1}{27} = \left(\frac{1}{3}\right)^{\square}$

b $\frac{64}{27} = \left(\frac{4}{3}\right)^{\square}$

c $\frac{27}{8} = \left(\frac{3}{\square}\right)^3$

d $\frac{\square}{16} = \left(\frac{1}{4}\right)^2$

e $\frac{27}{8} = \left(\frac{3}{2}\right)^{\square}$

f $\frac{\square}{625} = \left(\frac{2}{5}\right)^4$

g $\frac{81}{100} = \left(\frac{\square}{10}\right)^2$

h $\frac{256}{\square} = \left(\frac{4}{5}\right)^4$

i $\frac{25}{9} = \left(\frac{5}{3}\right)^{\square}$

j $\frac{\square}{144} = \left(\frac{11}{12}\right)^2$

k $\frac{125}{27} = \left(\frac{\square}{3}\right)^3$

l $\frac{32}{729} = \left(\frac{2}{3}\right)^{\square}$

9 Evaluate the following expressions in fully simplified fraction:



a $\left(\frac{1}{3}\right)^2$

b $\left(\frac{3}{5}\right)^3$

c $\left(\frac{1}{2}\right)^5$

d $\left(\frac{7}{11}\right)^2$

e $\left(\frac{3}{8}\right)^3$

f $\left(\frac{12}{13}\right)^2$

g $\left(\frac{5}{9}\right)^3$

h $\left(\frac{2}{3}\right)^4$

i $\left(\frac{4}{5}\right)^6$

j $\left(\frac{2}{5}\right)^3$

k $\left(\frac{5}{7}\right)^2$

l $\left(\frac{10}{14}\right)^3$

10 State whether the following fractions are equal to $\left(\frac{1}{2}\right)^3$:

a $\frac{3}{6}$

b $\frac{1}{2^3}$

c $\frac{3}{2^3}$

d $\frac{1}{8}$